Variant Filtering Automation:

A quality and annotation guided tool for prioritising variants from Next Generation Sequencing (NGS) data

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Background & Introduction

- Cost-effective NGS enables large-scale variant discovery, including in under-studied populations
- Variant filtering & prioritisation remain major bottlenecks in genomic analysis
- Existing tools require heavy manual input, lack reproducibility, and perform poorly on diverse or joint-called WES/WGS cohorts
- Challenges are amplified in African datasets due to high genetic diversity and limited population-specific resources
- Need for a customisable, transparent, reproducible workflow integrating QC and functional annotations for variant filtering and prioritisation

Research Question, Aims & Objectives

Can a Python-based annotation- and quality-guided automated workflow reliably prioritise clinically relevant variants from joint-called WES data in African genomic datasets?

To improve and assess an already existing automated variant filtering workflow for joint-called WES data, and to promote reproducibility of results.

1. Understand the current implementat of the Python-based workflow and its limitations.
2. Efficiently Integrate variant-level QC metrics and functional annotations into a structured decision-making framework.
3. Evaluate the tools performance by comparing filtered variants to manually curated or benchmark datasets.
4. Assess the tool’s reproducibility and scalability across different WES datasets.

Resources, Skills & Expected Outcomes

Data-Driven Project

- Data already generated
- Available on the Wits cluster
- No labwork
- Datasets to be use:
  - DDD-Africa
  - AAPC

Bioinformatics Skills

- Bioinformatics techniques
- Python
- Jupyter Notebooks
- Linux/Unix
- Command-Line Interface

Variant Filtering

- Online resource
- Publication

Relevance to Genomic Medicine in Africa

Africa remains underrepresented in global genomic research despite its rich genetic diversity, which holds crucial insights for understanding disease mechanisms and improving precision medicine.

The lack of computationally efficient and context-aware bioinformatics workflows hinders the variant interpretation and clinical translation.

This project will contribute to capacity building in genomic data analysis and strengthen bioinformatics expertise and infrastructure for Genomic Medicine in Africa.

Key Reading:

Pedersen, B. S. & Quinlan, A. R. (2021). Effective variant filtering and expected candidate counts for Mendelian disease studies. Genome Medicine / PMC